Introduction to Lattice-based Cryptography

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Abstract

In recent years, the interest in lattice-based cryptography has been steadily increasing. One primary reason is, of course, that problems related to lattices and their derivatives are widely believed to be *hard even for quantum computers*. Furthermore, there are settings where the algebraic structure of lattices is natural: e.g., for fully-homomorphic encryption.

This brief seminar is devoted to technical introduction to lattice-based cryptography. We give formal definitions of lattices and introduce key related hardness assumptions such as SVP, CVP, and LWE. We show how to build public-key encryption (PKE) scheme and give an overview of basic ideas of modern cryptanalysis.

Contents

1 Lattices **2**
- 1.1 Definition . . . 2
- 1.2 Fundamental Measures . . . 3
  - 1.2.1 Fundamental Domain . . . 3
  - 1.2.2 Successive Minima . . . 4
- 1.3 Minkowski's Theorem . . . 5
- 1.4 Lattice Computational Problems . . . 5

2 Learning With Errors (LWE) **5**
- 2.1 Key Generation . . . 6
- 2.2 Search-LWE (SLWE) and Decisional-LWE (DLWE) Assumptions . . . 6
- 2.3 LWE-based Public-Key Encryption . . . 9
- 2.4 Relation to Lattices . . . 10
- 2.5 Variations of LWE . . . 11
  - 2.5.1 Ring-LWE . . . 11
  - 2.5.2 Module-LWE . . . 13

1 Lattices

Notation and Basic Notions

All vectors are written in bold (e.g., $\mathbf{x}$ or $\mathbf{y}$) and considered *column* vectors by default. All real vector spaces are Euclidean: that is, for every $\mathbf{x}, \mathbf{y} \in V \subseteq \mathbb{R}^d$, we define the inner product $\langle \cdot, \cdot \rangle$ and ℓ_2 norm $\| \cdot \|$ in a standard fashion:

$$\langle \mathbf{x}, \mathbf{y} \rangle \triangleq \sum_{i \in [d]} x_i y_i, \quad \|\mathbf{x}\| \triangleq \sqrt{\langle \mathbf{x}, \mathbf{x} \rangle},$$

where we use $[d]$ to denote the set $\{1, \dots, d\}$. Additionally, we use (d) to denote the ranging set using 0-indexed notation: $\{0, \dots, d-1\}$. Other norms will be introduced on the go when needed. Finally, we use notation $x \leftarrow\!\$\ \mathcal{X}$ to denote sampling from the distribution $\mathcal{X}$. If $\mathcal{X}$ is a set, the underlying distribution is uniform over $\mathcal{X}$.

1.1 Definition

We start our discussion with the definition of the lattice. Following [dBvW25], we provide two definitions: one is rather algebraic, while the other is more intuitive and practical. In fact, both definitions will be helpful since some facts are sometimes easier to prove using the first definition.

Definition 1.1. Let V be a real Euclidean vector space of finite dimension d. The following two definitions of **lattice** $\Lambda \subseteq V$ of rank $r \leq d$ are equivalent:

(i) Λ is a discrete additive subgroup of V (with $\Lambda \cong \mathbb{Z}^r$ as $\mathbb{Z}$-modules).

(ii) $\Lambda = \mathcal{L}(B) \triangleq \{B\mathbf{z} : \mathbf{z} \in \mathbb{Z}^r\}$ for some matrix $B \in \mathbb{R}^{d \times r}$ where its columns, further denoted as $\{\mathbf{b}_1, \dots, \mathbf{b}_r\} \subseteq \mathbb{R}^d$, are linearly independent over V.

While the second definition is clear, the first one requires some deciphering. Since we work in the Euclidean space, it makes sense to define what it means for the set $S \subseteq V$ to be *discrete*. Intuitively, it should mean that any two points $\mathbf{s}_1, \mathbf{s}_2 \in S$ are separated by some "considerable" distance. To translate it into the mathematical language, we require that:

$$\inf_{\mathbf{s}_1 \neq \mathbf{s}_2 \in S} \|\mathbf{s}_1 - \mathbf{s}_2\| > 0.$$

Besides Λ being discrete, additive subgroup structure of Λ guarantees that Λ is closed under any $\mathbb{Z}$-linear combinations: that is, for any $\mathbf{x}, \mathbf{y} \in \Lambda$ and $\lambda, \mu \in \mathbb{Z}$, we have $\lambda\mathbf{x} + \mu\mathbf{y} \in \Lambda$. Showing this property might sometimes be easier than finding the matrix B in definition *(ii)* explicitly.

Proof of definition equivalence. $(ii) \Rightarrow (i)$. Clearly we have an additive subgroup structure. Discreteness comes from the fact that since B has full column rank, there exists some γ such that $\|B\mathbf{z}\| \geq \gamma \|\mathbf{z}\|$ (in fact, γ is the smallest singular value) for all $\mathbf{z} \in \mathbb{R}^r$. Thus, for $\mathbf{z} \in \mathbb{Z}^r \setminus \{0\}$, one has $\|B\mathbf{z}\| \geq \gamma > 0$.

$(i) \Rightarrow (ii)$. Assume $\Lambda \cong \mathbb{Z}^r$ is a discrete additive subgroup of V. Since Λ is free abelian of rank r, there exist elements $\mathbf{b}_1, \dots, \mathbf{b}_r \in \Lambda$ such that any $\lambda \in \Lambda$ can be written as $\lambda = \sum_{i \in [r]} z_i \mathbf{b}_i$, $z_i \in \mathbb{Z}$. Now we shall prove linear independence of $(\mathbf{b}_1, \dots, \mathbf{b}_r)$ over $\mathbb{R}$. In that is the case, then $\Lambda = \mathcal{L}(B)$ and we are done.

Assume otherwise that there is a non-trivial linear combination $\sum_{i \in [r]} \alpha_i \mathbf{b}_i = 0$. Pick $\{q_i^{(n)}\}_{n \geq 0}$ that approximate corresponding α_i ($\lim_{n \to \infty} q_i^{(n)} = \alpha_i$) and pick N_n be the product of all denominators of $\{q_i^{(n)}\}_{i \in [r]}$. Then, define $\lambda_n := \sum_{i \in [r]} N_n q_i^{(n)} \mathbf{b}_i \in \Lambda$. Now note that $\lim_{n \to \infty} \lambda_n = 0$, contradicting the discreteness.

Geometric Intuition. While definition *(i)* is useful, the definition *(ii)* gives an intuitive way of thinking of lattice $\Lambda = \mathcal{L}(B)$. If vector space V is typically thought as a set of all $\mathbb{R}$-linear combinations* $\sum_{i\in[d]} \mathbf{e}_i\mathbb{R}$ over some basis $\{\mathbf{e}_i\}_{i\in[d]}$, the lattice Λ can be thought as all $\mathbb{Z}$-linear combinations $\sum_{i\in[r]} \mathbf{b}_i\mathbb{Z}$ over real basis $\{\mathbf{b}_i\}_{i\in[r]} \subseteq \mathbb{R}^d$. See Figure 1.1 for illustration.

Remark 1.2. In this section we consider *full-rank* lattices only: that is, $\Lambda \cong \mathbb{Z}^d$ for $d = \dim(V)$.

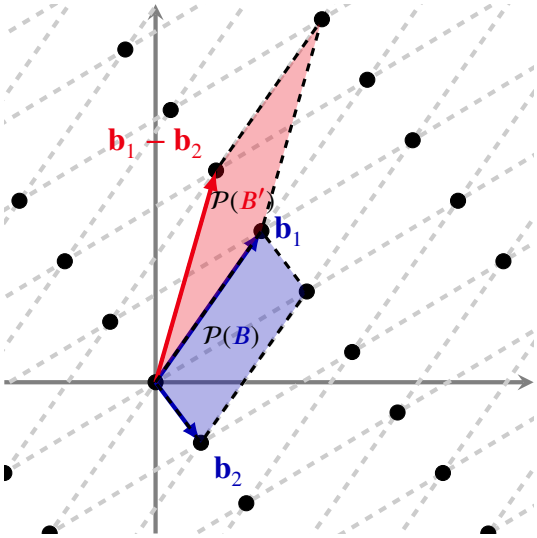


Figure 1.1: Lattice $\Lambda = \mathcal{L}(B)$ of rank 2 with basis $B = \{\mathbf{b}_1, \mathbf{b}_2\}$. B is an example of a "good" basis where basis vectors are almost orthogonal. The basis $B' = \{\mathbf{b}_1, \mathbf{b}_1 - \mathbf{b}_2\}$ of the same lattice $\Lambda = \mathcal{L}(B')$ is "worse" (the vectors seem more parallel and fundamental domain $\mathcal{P}(B')$ is skewed).

1.2 Fundamental Measures

1.2.1 Fundamental Domain

A basis B of $\Lambda = \mathcal{L}(B)$ is associated with the fundamental domain $\mathcal{P}(B) \triangleq \sum_{i\in[d]} \mathbf{b}_i[0,1)$†.

What is special about this parallelepiped is that it represents all elements in the quotient group V/Λ: any element $\mathbf{x} \in V$ can be uniquely written as $\mathbf{x} = \mathbf{v} + \mathbf{p}$ where $\mathbf{v} \in \Lambda$ and $\mathbf{p} \in \mathcal{P}(B)$. Moreover, the shape of this $\mathcal{P}(B)$ determines the "quality" of the basis B: if $\mathcal{P}(B)$ is skewed and is not concentrated in the origin, it is typically considered "bad" as some computational problems become much harder to solve. In turn, a non-skewed basis B' primarily concentrated around the origin is considered "good" as $\{\mathbf{b}'_i\}_{i\in[d]}$ much more closely resembles the orthogonal basis than one of B (and of course, the orthogonal basis in a sense is "perfect" because it is very "nice" to work with). Again, see Figure 1.1 for illustration.

Another natural quantity to consider is the volume of such parallelepiped, called *covolume*.

Definition 1.3 (Covolume)**.** The **covolume** $\mathrm{Vol}(\Lambda)$ of lattice Λ is defined as:

$$\mathrm{Vol}(\Lambda) \triangleq \mathrm{Vol}(V/\Lambda) = \sqrt{|\det(B^\top B)|}.$$

*We further use notation $S + S' := \{s + s' : s \in S, s' \in S'\}$ and for $a \in S$, set $aS := \{as : s \in S\}$.

†Sometimes, the interval $[-1/2, 1/2)$ is used instead of $[0,1)$ to center $\mathcal{P}(B)$ with respect to origin: in such case, $\mathcal{P}(B)$ is called a *centered parallelepiped of B*

Note that the definition of $\mathcal{P}(B)$ depends on the choice of B since the basis of Λ is generally not unique. Thus, it might seem that covolume should also depend on the choice of B. However, we show the following.

Proposition 1.4. The **covolume** $\mathrm{Vol}(\Lambda)$ is an invariant of Λ: that is, it does not depend on the choice of B.

Proof. Suppose $\Lambda = \mathcal{L}(B_0) = \mathcal{L}(B_1)$. Since B_0 and B_1 both generate Λ, one has $B_1 = U B_0$ for some matrix $U \in \mathrm{GL}_d(\mathbb{Z})$. Since U is an integer matrix with integer inverse, it is unimodular: that is, $|\det U| = 1$. It is then easy to see $|\det B_1^\top B_1| = |\det B_0^\top U^\top U B_0| = |\det B_0^\top B_0| = \mathrm{Vol}(\Lambda)^2$. □

1.2.2 Successive Minima

Turns out that one of the hardest computational problems is computing the shortest vectors of Λ. However, we need some preliminary work to formalize this statement.

Denote $B_\rho(\mathbf{x}_0) := \{\mathbf{x} \in V : \|\mathbf{x} - \mathbf{x}_0\| < \rho\}$ — an open ball in V of radius ρ centered at $\mathbf{x}_0$.

Definition 1.5 (Successive minima)**.** The ith **minimum** $\lambda_i(\Lambda)$ is the radius of the smallest sphere centered at the origin containing i linearly independent lattice vectors:

$$\lambda_i(\Lambda) \triangleq \inf\{\rho \in \mathbb{R}_{\geq 0} : \dim(\mathrm{span}(\Lambda \cap B_\rho(\mathbf{0}))) \geq i\}.$$

In particular, $\lambda_1(\Lambda) = \min_{\mathbf{x}\neq\mathbf{y}\in\Lambda} \|\mathbf{x} - \mathbf{y}\| = \min_{\mathbf{x}\in\Lambda\setminus\{\mathbf{0}\}} \|\mathbf{x}\|$ is called the *first minimum* while the vector of length $\lambda_1(\Lambda)$ — the shortest vector. The natural question to ask though is whether it exists at all: after all, in $\mathbb{R}^d$, for instance, it is clearly undefined. We give affirmative answer.

We shall further denote the result of Gram-Schmidt orthogonalization of B by B^*. The question that might naturally arise: given such orthogonalization, can't we then always turn a "bad" basis of lattice Λ into a "good" basis? Unfortunately, typically $\mathcal{L}(B) \neq \mathcal{L}(B^*)$: the orthogonalization of a lattice basis is (typically) not a lattice basis. However, the idea of Gram-Schmidt orthogonalization lies in the core of many attacks on lattices.

Proposition 1.6. Suppose $\Lambda = \mathcal{L}(B)$ is a lattice and B^* is the Gram-Schmidt orthogonalization of B with columns $\{\mathbf{b}_j^*\}_{j\in[d]}$. Then $\lambda_1(\Lambda) \geq \min_{j\in[d]} \|\mathbf{b}_j^*\| > 0$.

Proof (adapted from [MG02]). Consider a generic non-zero lattice vector $B\mathbf{x}$ ($\mathbf{x} \in \mathbb{Z}^d \setminus \{\mathbf{0}\}$) and let i be the biggest index such that $x_i \neq 0$. We first prove that $|\langle B\mathbf{x}, \mathbf{b}_i^*\rangle| \geq \|\mathbf{b}_i^*\|^2$. From the definition of i, we know $B\mathbf{x} = \sum_{j=1}^{i} \mathbf{b}_j x_j$. Thus:

$$\begin{aligned}
\langle B\mathbf{x}, \mathbf{b}_i^*\rangle = \sum_{j=1}^{i} \langle \mathbf{b}_j, \mathbf{b}_i^*\rangle x_j &= \langle \mathbf{b}_i, \mathbf{b}_i^*\rangle x_i \\
&= \langle \mathbf{b}_i^* + \sum_{j<i} \mu_{i,j}\mathbf{b}_j^*, \mathbf{b}_i^*\rangle x_i \\
&= \langle \mathbf{b}_i^*, \mathbf{b}_i^*\rangle x_i + \sum_{j<i} \mu_{i,j}\langle \mathbf{b}_j^*, \mathbf{b}_i^*\rangle x_i \\
&= \|\mathbf{b}_i^*\|^2 x_i.
\end{aligned}$$

From this follows $|\langle B\mathbf{x}, \mathbf{b}_i^*\rangle| \geq \|\mathbf{b}_i^*\|^2 |x_i| \geq \|\mathbf{b}_i^*\|^2$ and therefore $\|B\mathbf{x}\| \cdot \|\mathbf{b}_i^*\| \geq \|\mathbf{b}_i^*\|^2$. Clearly $\|\mathbf{b}_i^*\| \neq 0$ since $\mathbf{b}_i$'s are linearly independent, so $\|B\mathbf{x}\| \geq \|\mathbf{b}_i^*\| \geq \min_{j\in[d]} \|\mathbf{b}_j^*\|$. By definition of

$\lambda_1(\Lambda)$, the result follows. □

Proposition 1.7. For lattice Λ, there exists $\mathbf{v} \in \Lambda$ such that $\|\mathbf{v}\| = \lambda_1(\Lambda)$.

Proof. By the definition of $\lambda_1(\Lambda)$, there exists a sequence $\{\mathbf{v}_n\}_{n\in\mathbb{N}} \subseteq \Lambda$ such that $\lim_{n\to\infty} \|\mathbf{v}_n\| = \lambda_1(\Lambda)$. From Proposition 1.6, $\lambda_1(\Lambda) > 0$, so for all sufficiently large n it must be $\|\mathbf{v}_n\| \leq 2\lambda_1(\Lambda)$, i.e., lattice vector $\mathbf{v}_n$ belongs to the closed ball $\overline{B}_{2\lambda_1(\Lambda)}(\mathbf{0})$. But this set is compact, so we might extract a convergent subsequence $\{\mathbf{v}_{i_j}\}_{j\in\mathbb{N}}$ with limit $\mathbf{w} = \lim_{j\to\infty} \mathbf{v}_{i_j}$ and $\|\mathbf{w}\| = \lambda_1(\Lambda)$. We claim that $\mathbf{w} \in \Lambda$. By definition of $\mathbf{w}$, $\lim_{j\to\infty} \|\mathbf{v}_{i_j} - \mathbf{w}\| = 0$. Therefore for all sufficiently large j, $\|\mathbf{v}_{i_j} - \mathbf{w}\| < \lambda_1(\Lambda)/2$. By triangle inequality for sufficiently large j and all $k > j$, $\|\mathbf{v}_{i_j} - \mathbf{v}_{i_k}\| \leq \|\mathbf{v}_{i_j} - \mathbf{w}\| + \|\mathbf{w} - \mathbf{v}_{i_k}\| < \lambda_1(\Lambda)$. But $\mathbf{v}_{i_j} - \mathbf{v}_{i_k} \in \Lambda$, and no non-zero lattice vector can have length strictly less than $\lambda_1(\Lambda)$. Thus, $\mathbf{v}_{i_k} = \mathbf{v}_{i_j}$ for all $k > j$. Thus $\mathbf{w} = \mathbf{v}_{i_j} \in \Lambda$. □

1.3 Minkowski's Theorem

Clearly, one expects the dependence between the covolume of a lattice $\mathrm{Vol}(\Lambda)$ and the first minimum $\lambda_1(\Lambda)$. In particular, for smaller covolume we expect the smaller first minimum $\lambda_1(\Lambda)$ (or alternatively, one can say the lattice with large covolume cannot have a small first minimum). Moreover, by dimensionality analysis one should expect that $\lambda_1(\Lambda) \propto \mathrm{Vol}(\Lambda)^{1/d}$. The Minkowski theorem gives the inequality between these two quantities.

Theorem 1.8 (Minkowski's Theorems)**.** Let $\Lambda = \mathcal{L}(B)$ be the lattice of rank d with covolume $\mathrm{Vol}(\Lambda) = \det B$. Then,

First Minkowski's Theorem $\lambda_1(\Lambda) < \sqrt{d}\mathrm{Vol}(\Lambda)^{1/d}$.

Second Minkowski's Theorem $\prod_{i\in[d]} \lambda_i^{1/d} < \sqrt{d}\mathrm{Vol}(\Lambda)^{1/d}$.

Proof. See [MG02, Section 1.3].

While the Minkowski's Theorem turns out to be asymptotically tight for "bad" lattices, in general $\lambda_1(\Lambda)$ might be much smaller than $\sqrt{d}\mathrm{Vol}(\Lambda)^{1/d}$. Consider a very simple example: let $d = 2$ and $\mathbf{b}_1 := \varepsilon\mathbf{e}_1$, $\mathbf{b}_2 := (1/\varepsilon)\mathbf{e}_2$ (where $\mathbf{e}_1, \mathbf{e}_2$ is the canonical basis for $\mathbb{R}^2$). Minkowski's Theorem gives $\lambda_1(\Lambda) \leq \sqrt{2}$, however one easily sees that $\lambda_1(\Lambda) = \varepsilon$ can be made arbitrarily small.

Moreover, Minkowski's Theorem proof is not constructive: we know that a vector of length below $\sqrt{d}\mathrm{Vol}(\Lambda)^{1/d}$ exists, but there is no concrete algorithm to find it, let alone $\lambda_1(\Lambda)$. This became the core of one of the most well-known lattice computational problem: finding shortest vector in Λ. We introduce this and related problems in the next subsection.

1.4 Lattice Computational Problems

In this section, we finally define some problems that are currently believed to be unsolvable in polynomial time (under reasonable choice of parameters, of course).

- Finding $\mathbf{v}$ such that $\|\mathbf{v}\| = \lambda_1(\Lambda)$ of $\Lambda = \mathcal{L}(B)$ from a random basis B.
- Finding $\mathbf{v}$ such that $\|\mathbf{v}\| \leq \gamma\lambda_1(\Lambda)$.
- Given $\mathbf{t} \hookleftarrow\$ \mathbb{Z}^\mu$, find some $\mathbf{v} \in \mathcal{L}$ such that $\mathrm{dist}(\mathbf{t}, \mathbf{v}) \leq \gamma \cdot \mathrm{dist}(\mathbf{t}, \mathcal{L})$.

2 Learning With Errors (LWE)

2.1 Key Generation

Fix some modulo $q > 1$ (further assume it is prime). Our primary goal is to build public key cryptography schemes in the lattice-fashionable way. Turns out that one of such approaches is to use **Learning With Errors** assumptions.

Fix secret key $\mathbf{s} \in \mathbb{Z}_q^n$ which we assume, for the moment, is sampled uniformly randomly from the corresponding set. The public key, corresponding to $\mathbf{s}$, will be a vector of perturbed random inner products with $\mathbf{s}$. Specifically, the public information consists of multiple pairs $(\mathbf{a}, b) \in \mathbb{Z}_q^n \times \mathbb{Z}_q$ such that $b = \langle \mathbf{a}, \mathbf{s} \rangle + e$ for an *error* $e \in \mathbb{Z}_q$, typically taken from the distribution that produces very small values of e: for instance, one might think of $e \in [-\eta, \eta] \pmod q$ for $\eta \approx 3$. Denote such distribution by χ.

This way, we can give an intermediate version of our key generation as follows:

$\mathsf{KeyGen}(1^\lambda) \to (\mathsf{pk}, \mathsf{sk})$**:**

1. $\mathbf{s} \leftarrow\!\$\ \mathbb{Z}_q^\lambda$; `// Generate secret`
2. **for** $i \in [\lambda]$:
3. $\quad \mathbf{a}_i \leftarrow\!\$\ \mathbb{Z}_q^\lambda$, $e \leftarrow\!\$\ \chi$;
4. $\quad b_i \leftarrow \langle \mathbf{a}_i, \mathbf{s} \rangle + e$; `// Generate random pertubation`
5. **return** $(\mathsf{pk} = \{(\mathbf{a}_i, b_i)\}_{i \in [\lambda]}, \mathsf{sk} = \mathbf{s})$.

It is then natural to batch all pertubations into a single matrix A and vector $\mathbf{b}$.[‡] This way, our updated key generation scheme is in Figure 2.1 (where by χ^λ we denote the distribution over $\mathbb{Z}_q^\lambda$ where each component is taken from χ).

$\mathsf{KeyGen}(1^\lambda) \to (\mathsf{pk}, \mathsf{sk})$:

1. $(\mathbf{s}, A) \leftarrow\!\$\ \mathbb{Z}_q^\lambda \times \mathbb{Z}_q^{\lambda \times \lambda}$;
2. $\mathbf{e} \leftarrow\!\$\ \chi^\lambda$;
3. $b \leftarrow A\mathbf{s} + \mathbf{e}$;
4. **return** $(\mathsf{pk} = (A, b), \mathsf{sk} = \mathbf{s})$.

Figure 2.1: LWE-based Key Generation Scheme

Note that we of course expect that solving $A\mathbf{s} + \mathbf{e} = \mathbf{b}$ is hard with respect to $\mathbf{s}$. It should also be clear why we introduce the error term $\mathbf{e}$: assuming $\mathbf{e} = \mathbf{0}$, the equation simplifies to $A\mathbf{s} = \mathbf{b}$. This is a simple system of linear equations which can be further dealt with by means of, say, Gaussian elimination, which has $O(\lambda^3)$ complexity. The assumption of hardness of solving this equation for $\mathbf{e} \neq \mathbf{0}$ is what we introduce and treat in the next sections.

2.2 Search-LWE (SLWE) and Decisional-LWE (DLWE) Assumptions

In this section we shall finally formalize main assumptions about learning with errors problem. First, introduce the LWE instance generation algorithm as specified in Figure 2.2 below.

[‡]We also note that A might not necessarily be (and is typically not) the square matrix, but just for the sake of exposition we for now assume $A \in \mathbb{Z}_q^{\lambda \times \lambda}$.

$\mathsf{LWEGen}_{n,m,q,\chi}(1^\lambda, \mathbf{s}) \to \mathbb{Z}_q^{m\times n} \times \mathbb{Z}_q^m$:

1. $A \leftarrow\$ \mathbb{Z}_q^{m\times n}$;
2. $\mathbf{e} \leftarrow\$ \chi$ (over $\mathbb{Z}_q^m$);
3. $\mathbf{b} \leftarrow A\mathbf{s} + \mathbf{e}$;
4. **return** $(A, \mathbf{b})$.

Figure 2.2: LWE Instance Generation

Now, informally, LWE assumption states that, given output of $\mathsf{LWEGen}_{n,m,q,\chi}(1^\lambda, \mathbf{s})$, it is impossible to determine $\mathbf{s}$ (*Search-LWE assumption*). Similarly to many assumptions in other cryptographic systems (such as Diffie-Hellman assumption), LWE also has the decisional analog: distinguish output of $\mathsf{LWEGen}_{n,m,q,\chi}(1^\lambda, \mathbf{s})$ from the uniform sample over $\mathbb{Z}_q^{m\times n} \times \mathbb{Z}_q^m$ (*Decisional-LWE assumption*). We state both assumptions more formally below.

Definition 2.1 (Search-LWE Assumption)**.** For the given adversary $\mathcal{A}$ and LWE parameters (n, m, q, χ), define $\underline{\mathsf{Game\ SLWE}}^{\mathcal{A}}_{\mathsf{LWEGen}}$ as specified in Figure 2.3. Define the advantage of $\mathcal{A}$ as

$$\mathsf{Adv}^{\mathrm{SLWE}}_{\mathcal{A},\mathsf{LWEGen}}(\lambda) = \Pr\left[\underline{\mathsf{Game\ SLWE}}^{\mathcal{A}}_{\mathsf{LWEGen}}(\lambda) = 1\right].$$

We say that **Search-LWE assumption** (SLWE) holds if $\mathsf{Adv}^{\mathrm{SLWE}}_{\mathcal{A},\mathsf{LWEGen}}(\lambda) = \mathrm{negl}(\lambda)$.

Definition 2.2 (Decisional-LWE Assumption)**.** For the given adversary $\mathcal{A}$ and LWE parameters (n, m, q, χ), define $\underline{\mathsf{Game\ DLWE}}^{\mathcal{A}}_{\mathsf{LWEGen}}$ as specified in Figure 2.4. Define the advantage of $\mathcal{A}$ as

$$\mathsf{Adv}^{\mathrm{DLWE}}_{\mathcal{A},\mathsf{LWEGen}}(\lambda) = \left|\Pr\left[\underline{\mathsf{Game\ DLWE}}^{\mathcal{A}}_{\mathsf{LWEGen}}(\lambda) = 1\right] - \frac{1}{2}\right|.$$

We say that **Decisional-LWE assumption** (DLWE) holds if $\mathsf{Adv}^{\mathrm{DLWE}}_{\mathcal{A},\mathsf{LWEGen}}(\lambda) = \mathrm{negl}(\lambda)$ for all PPT adversaries $\mathcal{A}$.

$\underline{\mathsf{Game\ SLWE}}^{\mathcal{A}}_{\mathsf{LWEGen}}$:

1. $\mathbf{s} \leftarrow\$ \mathbb{Z}_q^n$;
2. $(A, \mathbf{b}) \leftarrow\$ \mathsf{LWEGen}_{n,m,q,\chi}(\mathbf{s})$;
3. $\widehat{\mathbf{s}} \leftarrow \mathcal{A}(A, \mathbf{b})$;
4. **return** $\mathbf{s} =_? \widehat{\mathbf{s}}$.

Figure 2.3: Search-LWE Game

$\underline{\mathsf{Game\ DLWE}}^{\mathcal{A}}_{\mathsf{LWEGen}}$:

1. $\mathbf{s} \leftarrow\$ \mathbb{Z}_q^n$;
2. $(A_0, \mathbf{b}_0) \leftarrow\$ \mathsf{LWEGen}_{n,m,q,\chi}(\mathbf{s})$;
3. $(A_1, \mathbf{b}_1) \leftarrow\$ \mathbb{Z}_q^{m\times n} \times \mathbb{Z}_q^m$;
4. $c \leftarrow\$ \{0, 1\}$;
5. $\widehat{c} \leftarrow \mathcal{A}(A_c, \mathbf{b}_c)$;
6. **return** $c =_? \widehat{c}$.

Figure 2.4: Decision-LWE Game

The natural question to ask is whether SLWE and DLWE assumptions are equivalent or not. In

fact, [Reg09] has proven that these notions are in fact equivalent in case $q = \mathsf{poly}(n)$. We show this in the following proposition.

Proposition 2.3 (Following [Reg09])**.** If $q = \mathsf{poly}(n)$, SLWE and DLWE assumptions are polynomially equivalent. To put it the other way, if the SLWE assumption holds for LWE instance with parameters $(n, m, q = \mathsf{poly}(n), \chi)$, then the DLWE assumption also holds.

Proof. SLWE $\Rightarrow$ DLWE reduction is obvious. We therefore show DLWE $\Rightarrow$ SLWE. Suppose exists adversary $\mathcal{A}$ that wins $\underline{\mathsf{Game\ DLWE}}^{\mathcal{A}}_{\mathsf{LWEGen}}$ in a non-negligible probability. We construct $\mathcal{B}^{\mathcal{A}}$ that wins $\underline{\mathsf{Game\ SLWE}}^{\mathcal{B}}_{\mathsf{LWEGen}}$ is a non-negligible probability as well.

The informal idea is as follows: suppose $\mathcal{B}$ wants to solve $A\mathbf{s} + \mathbf{e} = \mathbf{b}$. Turns out that $\mathcal{B}$ can guess the coordinates of $\mathbf{s}$ one at a time, starting from s_1. Suppose $\mathcal{B}$ picks some $\widetilde{s} \in \mathbb{Z}_q$ and wants to check whether $\widetilde{s} = s_1$. Select some $\boldsymbol{\Delta} \leftarrow\!\$\ \mathbb{Z}_q^m$ and let $\widetilde{A} = (\widetilde{\mathbf{a}}_1, \mathbf{a}_2, \ldots, \mathbf{a}_n)$ where $\widetilde{\mathbf{a}}_1 = \mathbf{a}_1 + \boldsymbol{\Delta}$ (and where the initial matrix is given by $A = (\mathbf{a}_1, \ldots, \mathbf{a}_n)$) and set $\widetilde{\mathbf{b}} := \mathbf{b} + \widetilde{s}\boldsymbol{\Delta}$. Now, if $\widetilde{s} = s_1$, then $\widetilde{\mathbf{b}} = \widetilde{A}\mathbf{s} + \mathbf{e}$, otherwise $\widetilde{\mathbf{b}} = \widetilde{A}\mathbf{s} + \mathbf{e} + (\widetilde{s} - s_1)\boldsymbol{\Delta}$. Since $\boldsymbol{\Delta}$ is uniform, it follows that in the latter case $\widetilde{\mathbf{b}}$ is uniformly random and independent of $\widetilde{A}$. Thus, using $\mathcal{A}$, adversary $\mathcal{B}$ has an effective test for chosen $\widetilde{s}$, and thus $\mathcal{B}$ can range over all possible q values, which is polynomial in n.

This way, by using $\mathcal{B}^{\mathcal{A}}_{\mathrm{SLWE}}$ defined in Figure 2.5, we obtain:

$$\mathsf{Adv}^{\mathrm{DLWE}}_{\mathcal{A},\mathsf{LWEGen}}(\lambda) \leq qn \cdot \mathsf{Adv}^{\mathrm{SLWE}}_{\mathcal{B}^{\mathcal{A}},\mathsf{LWEGen}}(\lambda). \quad \square$$

Remark 2.4. Note that it might seem like the factor of qn is too large to state the equivalence of DLWE and SLWE (indeed, in the elliptic curve cryptography, one typically uses q exponential in the security parameter). However, in practical applications, q is typically not considerably large: for instance, in Kyber [BDK+18], $q = 7681$ and $n = 256$!
Also note that the brute-force complexity without the DLWE oracle is q^n, which is of course overwhelmingly large! (compared to qn).

$\mathcal{B}^{\mathcal{A}}_{\mathrm{SLWE}}(1^\lambda, A = (\mathbf{a}_1, \ldots, \mathbf{a}_n), \mathbf{b}) \to \mathbb{Z}_q^m$:

1. $\widehat{\mathbf{s}} \leftarrow \mathbf{0} \in \mathbb{Z}_q^m$; `// Prepare the output`
2. **for** $i \in [n]$:
3. **for** $\widetilde{s} \in \mathbb{Z}_q$:
4. $\boldsymbol{\Delta} \leftarrow\!\$\ \mathbb{Z}_q^m$; `// Generate random offset`
5. $\widetilde{A} \leftarrow (\mathbf{a}_1, \ldots, \mathbf{a}_{i-1}, \widetilde{\mathbf{a}}_i, \mathbf{a}_{i+1}, \ldots, \mathbf{a}_n)$, $\widetilde{\mathbf{a}}_i := \mathbf{a}_i + \boldsymbol{\Delta}$;
6. $\widetilde{\mathbf{b}} \leftarrow \mathbf{b} + \widetilde{s}\boldsymbol{\Delta}$; `// Set offset vector` $\widetilde{\mathbf{b}}$ `and matrix` $\widetilde{A}$
7. $c \leftarrow \mathcal{A}(\widetilde{A}, \widetilde{\mathbf{b}})$; `// Run DLWE oracle`
8. **if** $c = 0$, set $\widehat{s}_i \leftarrow \widetilde{s}$ and **break**;
9. **return** $\widehat{\mathbf{s}}$.

Figure 2.5: Algorithm for finding the solution to LWE instance given oracle to the DLWE.

2.3 LWE-based Public-Key Encryption

Finally, let us define the public-key encryption (PKE) scheme based on SLWE and DLWE assumptions! We remind the reader of the definition of a PKE scheme.

Definition 2.5 (Public-Key Encryption Scheme)**.** A **public-key encryption scheme** Π_{PKE} is a triplet of the following PPT algorithms:

- $\mathsf{KeyGen}(1^\lambda) \to (\mathsf{pk}, \mathsf{sk})$ returns the pair of public key and secret key, respectively.
- $\mathsf{Enc}(\mathsf{pk}, m) \to c$ takes the public key and message and produces the ciphertext c.
- $\mathsf{Dec}(\mathsf{sk}, c) \to m$ takes the secret key and ciphertext and restores the message m.

We require **correctness**:

$$\forall m \ : \ \Pr\left[\widehat{m} = m \;\middle|\; \begin{array}{l}(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^\lambda) \\ \widehat{m} \leftarrow \mathsf{Dec}(\mathsf{sk}, \mathsf{Enc}(\mathsf{pk}, m))\end{array}\right] \geq 1 - \mathrm{negl}(\lambda),$$

and **IND-CCA** and **IND-CPA** security. To not overwhelm the reader with the details, we will only focus on the IND-CPA security with the corresponding $\underline{\text{Game INDCPA}}^{\mathcal{A}}_{\Pi}$ defined in Figure 2.6. We naturally define advantage $\mathsf{Adv}^{\text{IND-CPA}}_{\mathcal{A},\Pi_{\text{PKE}}}(\lambda) := 2\left|\Pr[\underline{\text{Game INDCPA}}^{\mathcal{A}}_{\Pi_{\text{PKE}}} = 1] - \frac{1}{2}\right|$ and say that Π_{PKE} is **IND-CPA secure** if $\mathsf{Adv}^{\text{IND-CPA}}_{\mathcal{A},\Pi_{\text{PKE}}}(\lambda) = \mathrm{negl}(\lambda)$ for all PPT $\mathcal{A}$.

$\underline{\text{Game INDCPA}}^{\mathcal{A}}_{\Pi}(1^\lambda)$:

1. $(\mathsf{pk}, \mathsf{sk}) \leftarrow \mathsf{KeyGen}(1^\lambda)$;
2. $(m_0, m_1) \leftarrow \mathcal{A}(\mathsf{pk})$; `// Typically one requires` $|m_0| = |m_1|$
3. $b \leftarrow\!\$\ \{0,1\}$, $c_b \leftarrow \mathsf{Enc}(\mathsf{pk}, m_b)$;
4. $\widehat{b} \leftarrow \mathcal{A}(\mathsf{pk}, m_0, m_1, c_b)$;
5. **return** $\widehat{b} =_? b$.

Figure 2.6: IND-CPA Security Game. $\mathcal{A}$ might be allowed to perform steps 2–3 $\mathrm{poly}(\lambda)$ times.

Now, we define the following PKE scheme based on (n, m, q, χ) assumption where the message space is $\{0,1\}$. The main idea is that encryption scheme will generate the new LWE instance $(A', \mathbf{b}')$ by taking a "small" linear combination of publicly known $(A, \mathbf{b})$. By additionally adding $m\lceil q/2 \rfloor$ to $\mathbf{b}'$, we obtain the encryption (further denote by $\lceil \cdot \rfloor : \mathbb{Q} \to \mathbb{Z}$ the function that rounds a given rational number to the nearest integer). Upon decryption, the holder of the secret key $\mathbf{s}$ will be able to check whether $A'\mathbf{s} - \mathbf{b}'$ produces the small error. If the term $m\lceil q/2 \rfloor$ was added to form $\mathbf{b}'$, the resultant difference would be large, suggesting that the original bit was 1. Otherwise, decryption would output 0. Let us put this formally in the definition below.

Definition 2.6 (LWE-based PKE)**.** Based on LWE instance with parameters (n, m, q, χ), define the PKE scheme $\Pi^{\text{LWE}}_{\text{PKE}}$ over message space $\{0,1\}$ as follows: KeyGen is specified in Figure 2.1, Enc in Figure 2.7, and Dec in Figure 2.8.

Since we have to work with "small" and "large" elements of $\mathbb{Z}_q$, by $\|x\|_\infty$ for $x \in \mathbb{Z}_q$ we denote x if $x \leq \frac{q}{2}$, and $q - x$ otherwise (for example, naturally, $\|q - 3\|_\infty = 3$).

$\mathsf{Enc}(\mathsf{pk}, m \in \{0,1\}) \to (c \in \mathbb{Z}_q^n \times \mathbb{Z}_q)$:

1. $\Delta\mathbf{r}, \Delta\mathbf{z} \leftarrow\$\ \chi^m$, $\Delta z' \leftarrow\$\ \chi$;
2. $\widetilde{\mathbf{a}} \leftarrow A^\top \Delta\mathbf{r} + \Delta\mathbf{z}$;
3. $\widetilde{b} \leftarrow \mathbf{b}^\top \Delta\mathbf{r} + \Delta z' + m\lceil q/2 \rfloor$;
4. **return** $c = (\widetilde{\mathbf{a}}, \widetilde{b})$.

Figure 2.7: Encryption in $\Pi_{\mathrm{PKE}}^{\mathrm{LWE}}$

$\mathsf{Dec}(\mathsf{sk}, c) \to m$:

1. Parse $(\widetilde{\mathbf{a}}, \widetilde{b}) \leftarrow c$;
2. $\delta \leftarrow \widetilde{b} - \mathbf{s}^\top \widetilde{\mathbf{a}}$;
3. **return** 1 iff $\|\delta\|_\infty > q/4$.

Figure 2.8: Decryption in $\Pi_{\mathrm{PKE}}^{\mathrm{LWE}}$

Now we show that $\Pi_{\mathrm{PKE}}^{\mathrm{LWE}}$ is a correct and secure PKE scheme.

Proposition 2.7. $\Pi_{\mathrm{PKE}}^{\mathrm{LWE}}$ is correct and IND-CPA secure under DLWE assumption.

Proof. (*Correctness*) Let us first show correctness. Assume $\mathsf{sk} = \mathbf{s}$ and $\mathsf{pk} = (A, \mathbf{b})$ is the valid key-pair (that is, $A\mathbf{s} + \mathbf{e} = \mathbf{b}$ for "small" $\mathbf{e} \leftarrow\$\ \chi^m$) and we formed $(\widetilde{\mathbf{a}}, \widetilde{b})$ according to Figure 2.7. Let us try to decode this cipher by means of Figure 2.8:

$$\begin{aligned}\delta &= \mathbf{b}^\top \Delta\mathbf{r} + \Delta z' + m\lceil q/2 \rfloor - \mathbf{s}^\top A^\top \Delta\mathbf{r} - \mathbf{s}^\top \Delta\mathbf{z} \\ &= (\mathbf{b} - A\mathbf{s})^\top \Delta\mathbf{r} + (\Delta z' - \mathbf{s}^\top \Delta\mathbf{z}) + m\lceil q/2 \rfloor \\ &= (\mathbf{e}^\top \Delta\mathbf{r} - \mathbf{s}^\top \Delta\mathbf{z} + \Delta z') + m\lceil q/2 \rfloor\end{aligned}$$

Now note that the quantity marked in blue is small. Indeed, let us show why. Assume that χ produces elements of $\mathbb{Z}_q$ with the modulus not exceeding η. Then[§],

$$\|\mathbf{e}^\top \Delta\mathbf{r} - \mathbf{s}^\top \Delta\mathbf{z} + \Delta z'\|_\infty \le \|\mathbf{e}^\top \Delta\mathbf{r}\|_\infty + \|\mathbf{s}^\top \Delta\mathbf{z}\|_\infty + \|\Delta z'\|_\infty \le n\eta^2 + n\eta + \eta \le 2n\eta.$$

Clearly, $2n\eta$ is a small quantity (assuming $\eta \ll q/8n$). Now, if $m = 0$, we get that $\|\delta\|_\infty = \|(\text{blue part})\|_\infty < q/4$, so decryption works. If $m = 1$, then $\delta = \lceil q/2 \rfloor + (\text{blue part})$, which is a large value, so the decryption outputs 1, as expected.

(*IND-CPA Security*). Suppose $\mathcal{A}$ solves the $\underline{\mathsf{Game\ INDCPA}}_{\Pi}^{\mathcal{A}}$ with a non-negligible probability. We construct the $\mathcal{B}^{\mathcal{A}}$ that breaks the $\underline{\mathsf{Game\ DLWE}}_{\mathsf{LWEGen}}^{\mathcal{B}}$ using $\mathcal{A}$ as an oracle. The high-level idea of the proof is the following: suppose $\mathcal{B}$ is given a tuple $(A, \mathbf{b})$. If $(A, \mathbf{b})$ is a valid LWE instance, then it is a valid public key in $\Pi_{\mathrm{PKE}}^{\mathrm{LWE}}$, so $\mathcal{B}$ might run $\mathcal{A}$ on $\mathsf{pk}^* \leftarrow (A, \mathbf{b})$. Then, if $\mathcal{A}$ correctly guesses the bit, then likely pk^* was a valid key. In turn, if pk^* is a "garbage", with probability approximately $1/2$, the $\mathcal{A}$ would lose. Thus, $\mathcal{B}$ decision will be based on whether $\mathcal{A}$ guesses the correct bit or not. We encapsulate this in Figure 2.9. One can then show:

$$\mathsf{Adv}_{\mathcal{A},\Pi}^{\text{IND-CPA}}(\lambda) \le 2\mathsf{Adv}_{\mathcal{B}^{\mathcal{A}},\mathsf{LWEGen}}^{\mathrm{DLWE}}(\lambda). \quad \square$$

2.4 Relation to Lattices

The reasonable question that the reader might rightfully have at this point: where are the lattices? So far, we have spent the whole Section 1 describing lattices, while the current section does not explicitly use them in any way.

[§]In the sequel, we will show that $\mathbf{s}$ can be taken from the same distribution as the error term $\mathbf{e}$ (that is, from χ), making upper bound even smaller.

$\mathcal{B}^{\mathcal{A}}_{\mathrm{DLWE}}(1^\lambda, A, \mathbf{b}) \to \{0,1\}$:

1. pass $\mathsf{pk}^* \leftarrow (A, \mathbf{b})$ to $\mathcal{A}$;
2. get (m_0, m_1) from $\mathcal{A}$;
3. choose $d \leftarrow\$ \{0,1\}$;
4. send $c_d \leftarrow \mathsf{Enc}(\mathsf{pk}^*, m_d)$ to $\mathcal{A}$;
5. $\mathcal{A}$ responds with $\widehat{d}$;
6. **return** $d =_? \widehat{d}$.

Figure 2.9: Algorithm for finding the solution to DLWE instance given oracle to the IND-CPA.

In fact, turns out that the LWE problem can be viewed as the lattice problem!

Definition 2.8. For the LWE instance $(A, \mathbf{b})$ over parameters (n, m, q, χ), we define the associated lattice $\Lambda_q(A)$ as follows:

$$\Lambda_q(A) \triangleq \{\mathbf{y} \in \mathbb{Z}^m : \mathbf{y} = A\mathbf{x} \pmod q \text{ for some } \mathbf{x} \in \mathbb{Z}^n\} = A\mathbb{Z}^n + q\mathbb{Z}^m.$$

Now note that the vector $\mathbf{b}$ does not lie on the lattice $\Lambda_q(A)$ (unless $\mathbf{e} = \mathbf{0}$ which makes LWE instance unsecure). But note that due to the modular equation, $\mathbf{b} = A\mathbf{s} + q\mathbf{z} + \mathbf{e}$ for some $\mathbf{z} \in \mathbb{Z}^m$ and since $A\mathbf{s} + q\mathbf{z} \in \Lambda_q(A)$, the distance from $\mathbf{b}$ to $\Lambda_q(A)$ is bounded by $\|\mathbf{e}\|$. That said, although $\mathbf{b}$ is not in the lattice, it is very close to it! Therefore, almost surely, the closest vector to $\mathbf{b}$ is $A\mathbf{s} + q\mathbf{z}$. Now imagine that we know how to solve CVP defined in **??**. Then, based on $\mathbf{b}$, we can derive $A\mathbf{s} + q\mathbf{z}$, and then easily obtain $\mathbf{s}$ by solving a system of linear equations. Note that the approximation factor depends on the underlying distribution χ^m, from which $\mathbf{e}$ is sampled.

This way, lattices are crucial for cryptanalysis of LWE-based schemes.

2.5 Variations of LWE

Let us analyze the efficiency of the PKE scheme provided in Definition 2.6.

Proposition 2.9. In the PKE scheme $\Pi^{\mathrm{LWE}}_{\mathrm{PKE}}$ over LWE instance (n, m, q, χ), the public key size is $O(m \log q)$, secret key size is $O(n \log q)$, and the codeword size is $O(m \log q)$: therefore, sending a B-bit message takes $O(Bm \log q)$ space.

Remark 2.10. One might ask why the public key size is not $O(nm \log q)$, given that we need to store the matrix A. Fortunately, A is sampled randomly, so instead of storing the whole matrix, one might store the seed from which one can derive A instead.

Unfortunately, such key and ciphertext sizes are significantly larger than those of elliptic curve-based schemes (in fact, if λ is taken as a security parameter, then ciphertext size scales as $O(\lambda^2)$). This led to the natural question: can we reduce these sizes?

2.5.1 Ring-LWE

Turns out that one possible answer to this question is to use objects other than $\mathbb{Z}_q^n$ as elements that possess certain useful structure: for instance, the popular choice is to use $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$ for $\mathcal{R} = \mathbb{Z}[T]/(T^n + 1)$ and n being a power-of-two. In such case, one might think of $\mathcal{R}$ either by:

(a) $(n-1)$-degree polynomials with operations modulo $T^n + 1$;

(b) the ring of integers $\mathbb{Z}(\zeta_{2n})$ since $\mathcal{R} \cong \mathbb{Z}[T]/(\Phi_{2n}(T)) \cong \mathbb{Z}(\zeta_{2n})$ where $\zeta_N := e^{2\pi i/N}$.

Now, instead of having $(A, \mathbf{b}) \in \mathbb{Z}_q^{m\times n} \times \mathbb{Z}_q^m$ as a public key, we will store the public key as a pair $(a, b) \in \mathcal{R}_q^2$ where $b = as + e$, but over the polynomial space. However, we should clarify what "small" means in the context of $\mathcal{R}_q$ (since e here comes from some small distribution). For this reason, given $f(T) = \sum_{i\in(n)} f_i T^i \in \mathcal{R}_q$, define:

$$\|f\|_\infty \triangleq \max_{i\in(n)} \|f_i\|_\infty.$$

So in other words, polynomial $f(T) \in \mathcal{R}_q$ is large as long as its largest coefficient is large.

Further, we identify Ring-LWE instance by a tuple of $(\mathcal{R}, n, q, \chi)$, where (i) additive group of $\mathcal{R}$ is isomorphic to $\mathbb{Z}^n$, (ii) instead of $\mathbb{Z}_q^n$ in "plain" LWE, we use field $\mathcal{R}_q := \mathcal{R}/q\mathcal{R}$, and (iii) χ is a (small) random distribution over $\mathcal{R}_q$.

Proposition 2.11. Using Ring-LWE structure, we can reduce ciphertext size for a n-bit message of PKE Scheme from Definition 2.6 from $O(nm \log q)$ to $O(n \log q)$. Moreover, computing expression $as + e$ now takes only $O(n \log n)$ complexity instead of $O(nm)$.

The reason for $O(n \log n)$ complexity for computing $as + e$ comes from the fact that we can utilize Number Theoretic Transform (NTT/FFT) for fast polynomial multiplication. The reason for reduced ciphertext size is less evident, but the idea is the following: instead of adding a $m\lceil q/2 \rfloor$ term to the second ciphertext piece in Figure 2.7, we are going to represent n bits of message $(m_i)_{i\in(n)}$ as a polynomial $m(T) := \sum_{i\in(n)} m_i T^i \in \mathcal{R}_q$. We specify details in the definition below.

Definition 2.12. Based on Ring-LWE instance with parameters $(\mathcal{R}, n, q, \chi)$, define the PKE scheme $\Pi_{\text{PKE}}^{\text{RLWE}}$ over message space $\{0,1\}^n$ as follows: KeyGen is specified in Figure 2.10, Enc in Figure 2.11, and Dec in Figure 2.12.

KeyGen$(1^\lambda) \to$ (pk, sk):

1. $(s, a) \leftarrow\$ \mathcal{R}_q^2$;
2. $e \leftarrow\$ \chi$ (over $\mathcal{R}_q$);
3. $b \leftarrow as + e \in \mathcal{R}_q$;
4. **return** $((a, b), s)$.

Figure 2.10: Ring-LWE-based Key Generation

Enc(pk, $(m_i)_{i\in(n)} \in \{0,1\}^n$) $\to c$:

1. $m \leftarrow \sum_{i\in(n)} m_i T^i \in \mathcal{R}_q$;
2. $\Delta r, \Delta z, \Delta z' \leftarrow\$ \chi$;
3. $\widetilde{a} \leftarrow a\Delta r + \Delta z$;
4. $\widetilde{b} \leftarrow b\Delta r + \Delta z' + \lceil \frac{q}{2} \rfloor m$;
5. **return** $c = (\widetilde{a}, \widetilde{b})$.

Figure 2.11: Ring-LWE-based Encryption

Dec(sk, c) $\to m$:

1. Parse $(\widetilde{a}, \widetilde{b}) \leftarrow c$;
2. $\delta \leftarrow \widetilde{b} - s\widetilde{a}$;
3. **return** 1 iff $\|\delta\|_\infty > \frac{q}{4}$.

Figure 2.12: Ring-LWE-based Decryption

Another valid question is whether introducing such structure should worsen the security: so in other words, given appropriate Ring-LWE instance $(\mathcal{R}, n, q, \chi)$, is $\Pi_{\text{PKE}}^{\text{RLWE}}$ secure? Turns out that there are no algorithms for breaking analogous Ring-LWE-based assumption significantly faster than "plain" version of the LWE. To put this assumption formally, we slighly modify Definition 2.1.

Definition 2.13. For the given adversary $\mathcal{A}$ and Ring-LWE parameters $(\mathcal{R}, n, q, \chi)$, define $\underline{\text{Game SRLWE}}^{\mathcal{A}}$ as specified in Figure 2.13. Define the advantage of $\mathcal{A}$ as

$$\mathsf{Adv}^{\text{SRLWE}}_{\mathcal{A}}(\lambda) = \Pr\left[\underline{\text{Game SRLWE}}^{\mathcal{A}}(\lambda) = 1\right].$$

We say that **Search-Ring-LWE assumption** (SRLWE) holds if $\mathsf{Adv}^{\text{SRLWE}}_{\mathcal{A}}(\lambda) = \text{negl}(\lambda)$ for all PPT adversaries $\mathcal{A}$. Decisional version of Ring-LWE is defined similarly to Definition 2.2.

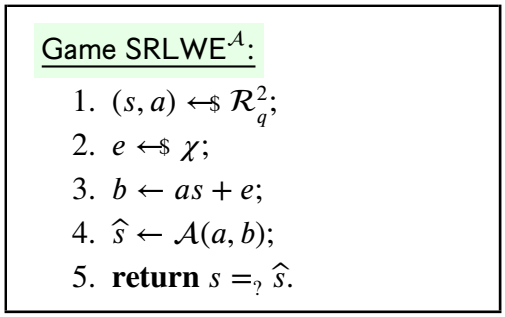


Figure 2.13: Search-Ring-LWE Game

We leave the following proposition without a proof as it closely resembles one specified in Proposition 2.7. For concrete details and survey on possible attacks, see for example [LLZ+18].

Proposition 2.14. $\Pi^{\text{RLWE}}_{\text{PKE}}$ is correct and IND-CPA secure under Decisional Ring-LWE assumption.

2.5.2 Module-LWE

The idea of Module-LWE is to in a way "blend" together Ring-LWE and plain-LWE. Namely, instead of using $\mathcal{R}_q$ in Ring-LWE, we will use vectors from $\mathcal{R}_q^\ell$. Note that equipping $\mathcal{R}_q$ with a vector structure naturally induces metric notions we have used before: e.g., for $\boldsymbol{f} = (f_1, \ldots, f_\ell) \in \mathcal{R}_q^\ell$, we have $\|\boldsymbol{f}\|_\infty := \max_{i\in[\ell]} \|f_i\|_\infty$ and $\langle \boldsymbol{f}, \boldsymbol{g}\rangle = \sum_{i\in[\ell]} f_i g_i$ is a valid inner product that produces output in $\mathcal{R}_q$.

The analogy to plain-LWE equation $\mathbf{b} = A\mathbf{s} + \mathbf{e}$ for $A \in \mathbb{Z}_q^{n\times m}$, $\mathbf{s} \in \mathbb{Z}_q^m$, and $\mathbf{b}, \mathbf{e} \in \mathbb{Z}_q^n$, is now $\mathbf{b}_\mathcal{R} = A_\mathcal{R}\mathbf{s}_\mathcal{R} + \mathbf{e}_\mathcal{R}$ for $A_\mathcal{R} \in \mathcal{R}_q^{k\times\ell}$, $\mathbf{s} \in \mathcal{R}_q^\ell$, and $\mathbf{s}_\mathcal{R}, \mathbf{e}_\mathcal{R} \in \mathcal{R}_q^k$, where operations are done inside ring $\mathcal{R}_q$. We call $(A, \mathbf{b})$ a Module-LWE instance with parameters $(\mathcal{R}, k, \ell, q, \chi)$ if it is an output of $\mathsf{MLWEGen}_{\mathcal{R},k,\ell,q,\chi}$ defined in Figure 2.14. Now we define Module-LWE assumption in the definition below.

Definition 2.15. For the given adversary $\mathcal{A}$ and Module-LWE parameters $(\mathcal{R}, k, \ell, q, \chi)$, define $\underline{\text{Game SMLWE}}^{\mathcal{A}}$ as specified in Figure 2.15. Define the advantage of $\mathcal{A}$ as

$$\mathsf{Adv}^{\text{SMLWE}}_{\mathcal{A}}(\lambda) = \Pr\left[\underline{\text{Game SMLWE}}^{\mathcal{A}}(\lambda) = 1\right].$$

We say that **Search-Module-LWE assumption** (SMLWE) holds if $\mathsf{Adv}^{\text{SMLWE}}_{\mathcal{A}}(\lambda) = \text{negl}(\lambda)$ for all PPT adversaries $\mathcal{A}$. Decisional Module-LWE is defined similarly to Definition 2.2.

At first glance, it might seem that Module-LWE looks much less efficient than Ring-LWE: for

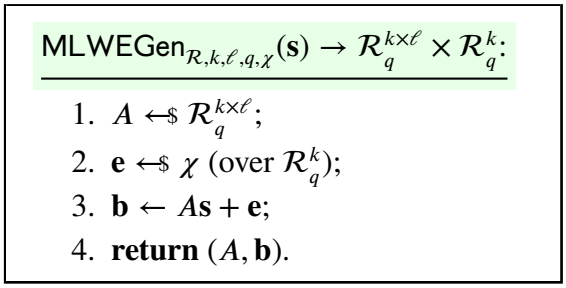


Figure 2.14: Modular-LWE $(\mathcal{R}, k, \ell, q, \chi)$ instance generation

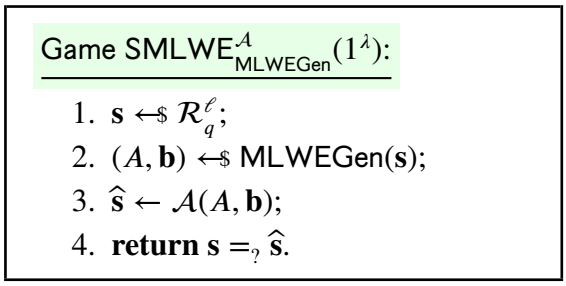


Figure 2.15: Search Modular-LWE Game

instance, why should representing A as a matrix $\mathcal{R}_q^{k\times\ell}$ be any more efficient than using a single $\mathcal{R}_q$? Notice that to adjust the security of Ring-LWE, one must regulate n — the power of cyclotomic polynomial. However, it must be a power of two, which makes such parameter adjusting highly problematic. For Module-LWE, one typically fixes (q, n) (which fully determine $\mathcal{R}_q$-arithmetic efficiency), and then changes ℓ to set target security level.

This way, key sizes (which are primarily determined by a storage of $\mathbf{b}$ by Remark 2.10) turn out to be more or less equal for both Ring-LWE and Module-LWE:

- *Dilithium* [DKL+18], being a Module-LWE, uses parameters $q = 2^{23} - 2^{13} + 1$, $n = 256$, and $(k, \ell) \in \{(3,2), (4,3), (5,4), (6,5)\}$; If we take, say $(k, \ell) = (4, 3)$, we need $kn = 1024$ elements of $\mathbb{Z}_q$ to store $\mathbf{b} \in \mathcal{R}_q^k$.
- *NewHope* [ADPS16], being a Ring-LWE, uses $q = 2^{13} + 2^{12} + 1$ and $n = 1024$. Notice that although we have a single polynomial $b \in \mathcal{R}_q$, we still need 1024 elements of $\mathbb{Z}_q$ for storage.

Turns out that implementing analogous PKE scheme to ones in Definition 2.6 and Definition 2.12 based on Module-LWE assumptions leads to the Kyber PKE description! We specify the details in **??**.

Acknowledgements

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